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(54) METHOD FOR PRODUCING SUBSTITUTED POLYCYCLIC PYRIDONE DERIVATIVE AND CRYSTAL OF SAME

VERFAHREN ZUR HERSTELLUNG EINES SUBSTITUIERTEN POLYCYCLISCHEN PYRIDONDERIVATS UND KRISTALL DAVON

MÉTHODES DE PRODUCTION DE DÉRIVÉS SUBSTITUÉS DE PYRIDONE POLYCYCLIQUE ET CRISTAL CORRESPONDANT

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- **Greene's protective groups in organic synthesis, pages 27-31 , 97- 99, 475-489, 503-507, ISBN: 978-1-118-05748-3**

Remarks:
The file contains technical information submitted after the application was filed and not included in this specification

EP 3 473 629 B1

Description

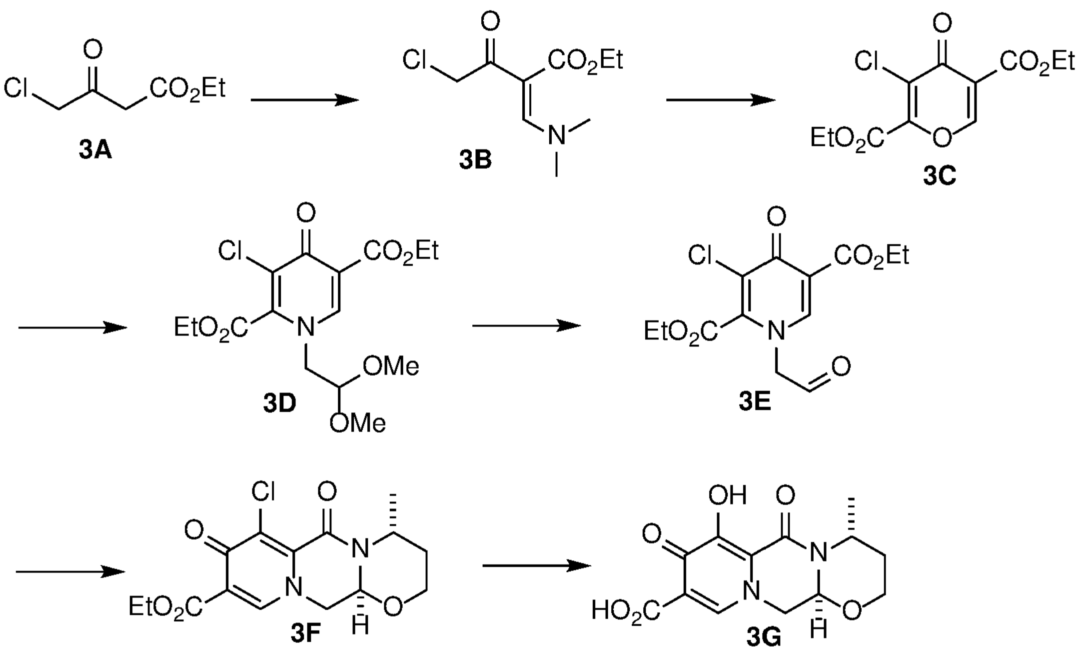
[Technical Field]

[0001] The present invention relates to a process for preparing substituted polycyclic pyridone derivatives and crystals thereof. Specifically, the present invention relates a process for preparing substituted polycyclic pyridone derivatives having cap-dependent endonuclease inhibitory activity and intermediates thereof.

[Background Art]

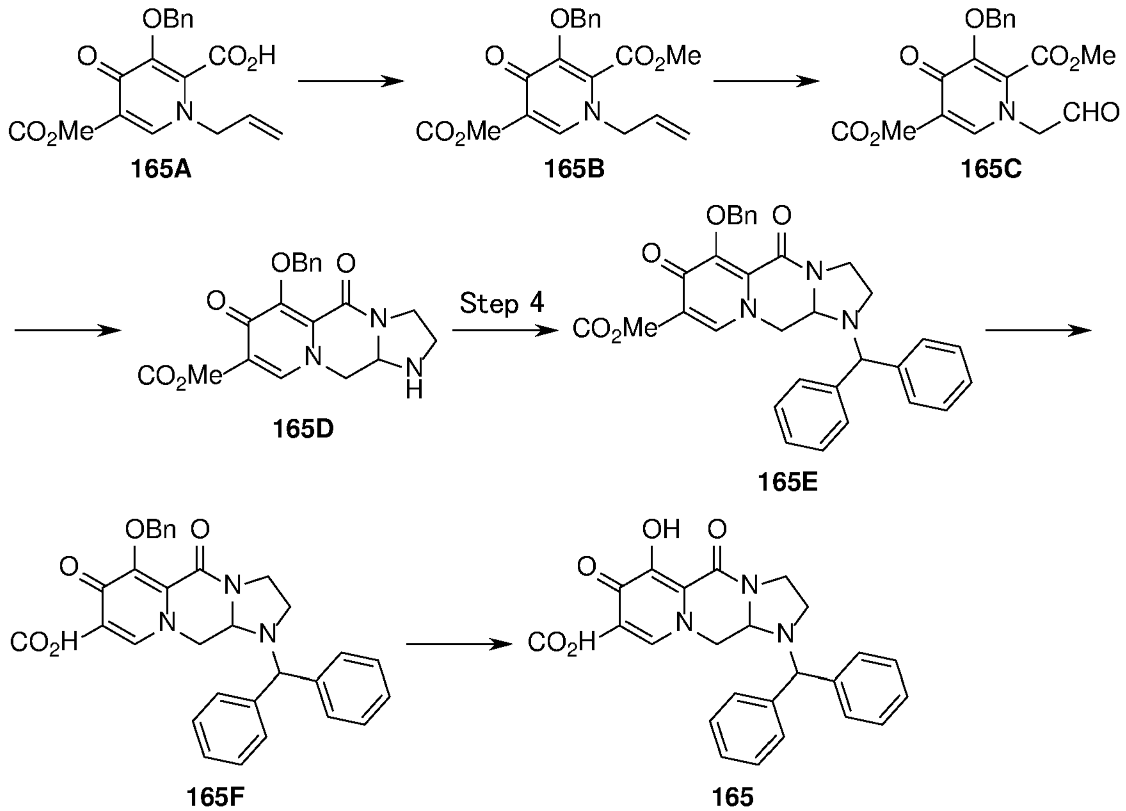
[0002] WO2010/110409 (Patent Document 1) discloses a process for preparing a polycyclic pyridone derivative using a pyrone derivative and a pyridone derivative (Example 3).

[Chem. 1]



[0003] WO2010/147068 (Patent Document 2) and WO2012/039414 (Patent Document 3) disclose a process using a pyridone derivative for the preparation of a polycyclic pyridone derivative (Example 165).

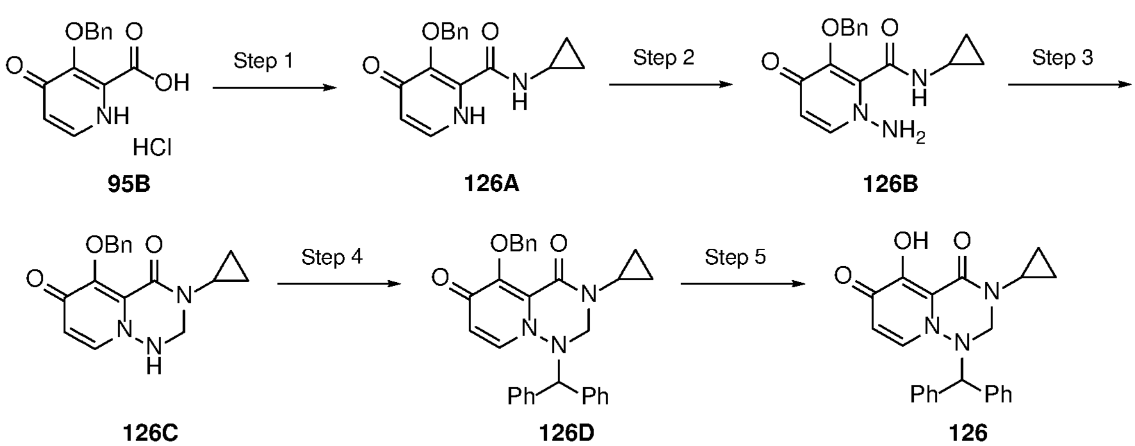
[Chem. 2]



[0004] However, Patent Documents 1 to 3 do not describe that use of a benzyl-protected polycyclic pyridone derivative in a coupling step of an optically active polycyclic pyridone derivative with a thiepin derivative reduce the optical purity of product. Also, there is neither description nor suggestion in the Patent Documents 1 to 3 that the coupling reaction proceeds in good yield without reduction in optical purity when the coupling reaction is carried out using a hexyl-protected polycyclic pyridone derivative. Furthermore, there is neither description nor suggestion that the reaction proceeds in high yield without reduction in optical purity when the reaction is carried out in the presence of a magnesium salt in the reaction to exchange the protecting group in the polycyclic pyridone derivative from a protecting group other than unsubstituted alkyl to unsubstituted alkyl.

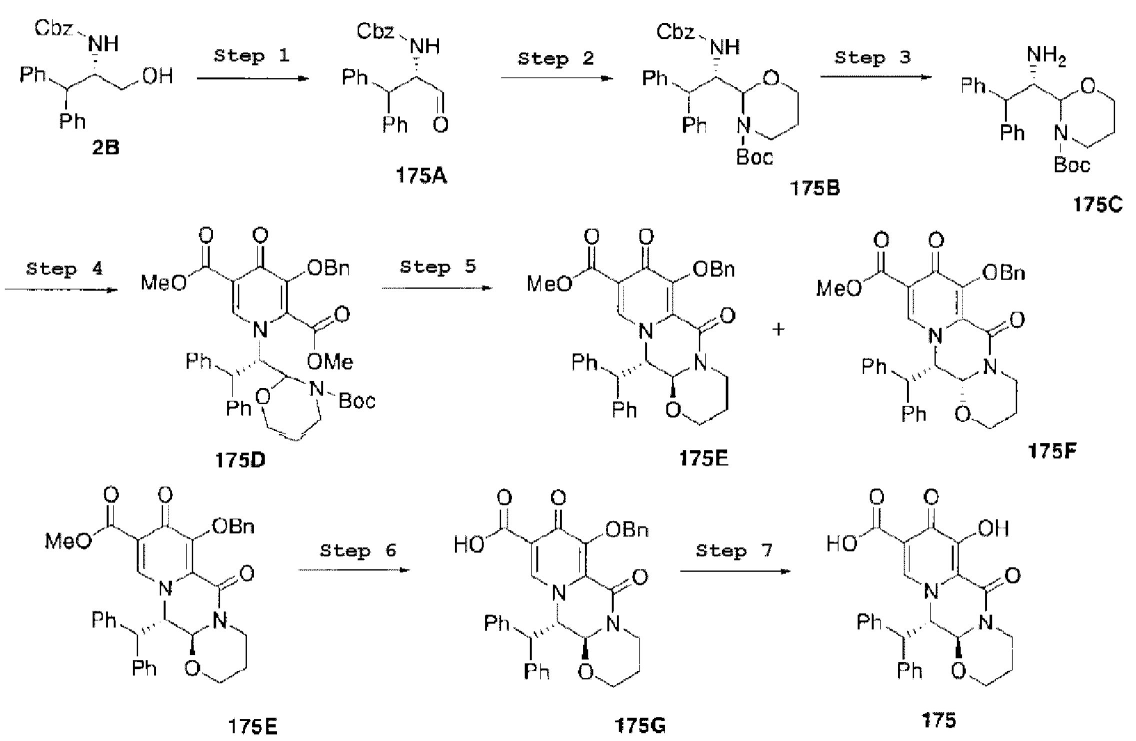
[0005] Patent Document 1 discloses the following process comprising a step of coupling a benzyl-protected polycyclic pyridone derivative with a benzhydryl derivative (Example 21). However, there is neither description nor suggestion of any step to exchange the protective group in the polycyclic pyridone derivative.

[Chem. 3]



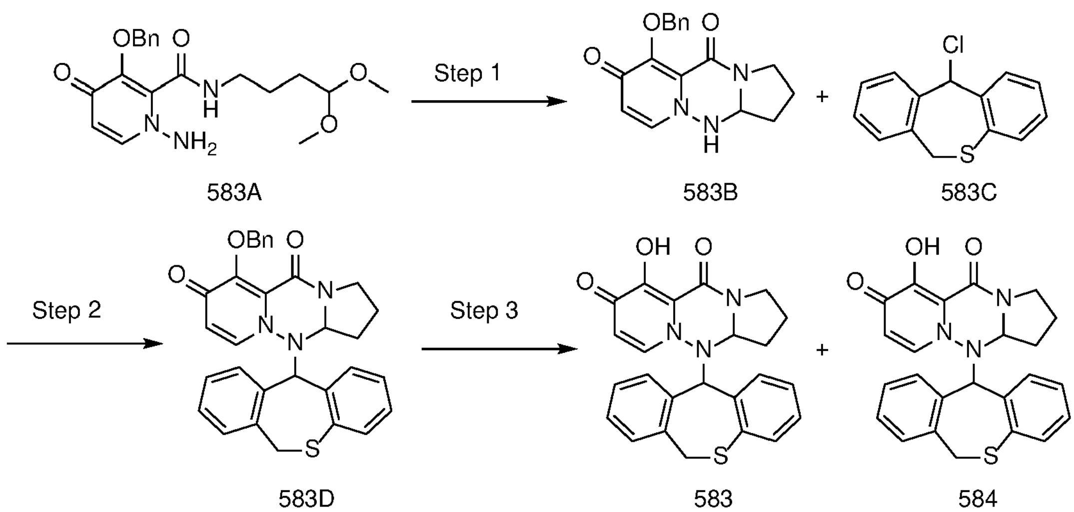
[0006] Patent Document 2 discloses a step of coupling a substituted tricyclic pyridone derivative with a benzhydryl derivative (Example 175). However, there is neither description nor suggestion of any step to exchange the protective group in the tricyclic pyridone derivative.

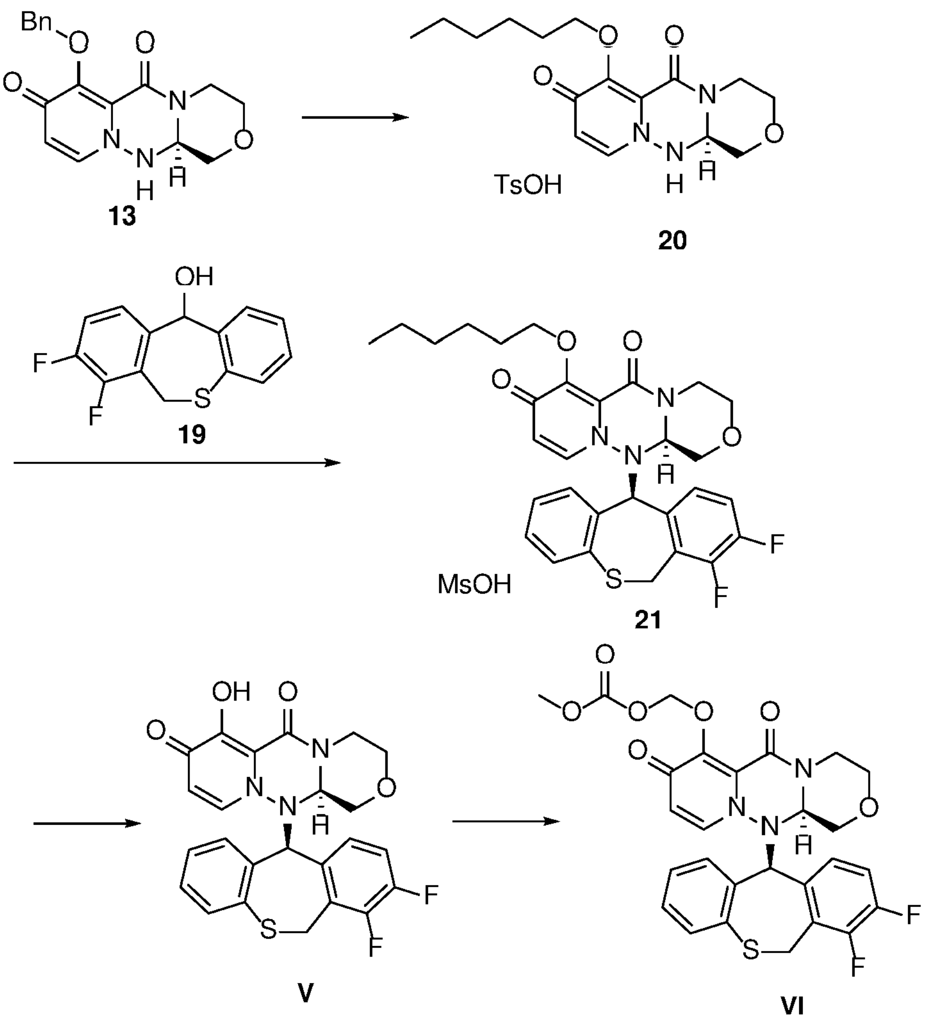
[Chem. 4]



[0007] Patent Document 2 discloses a step of coupling a substituted tricyclic pyridone derivative with a thiepin derivative (Examples 583 and 584). However, there is neither description nor suggestion of any step to exchange the protective group of the tricyclic pyridone derivative or reduction of the optical purity.

[Chem. 5]





Step 1-1: Compound 20

[0129] 1-Hexanol (22.5 g, 220 mmol) and THF (24.6 g) were combined, and the temperature of the mixture was adjusted to 20°C. A solution of isopropyl magnesium chloride in THF (2 mol/L, 7.2 g, 14.7 mmol) was added to the mixture to prepare a solution of magnesium hexoxide.

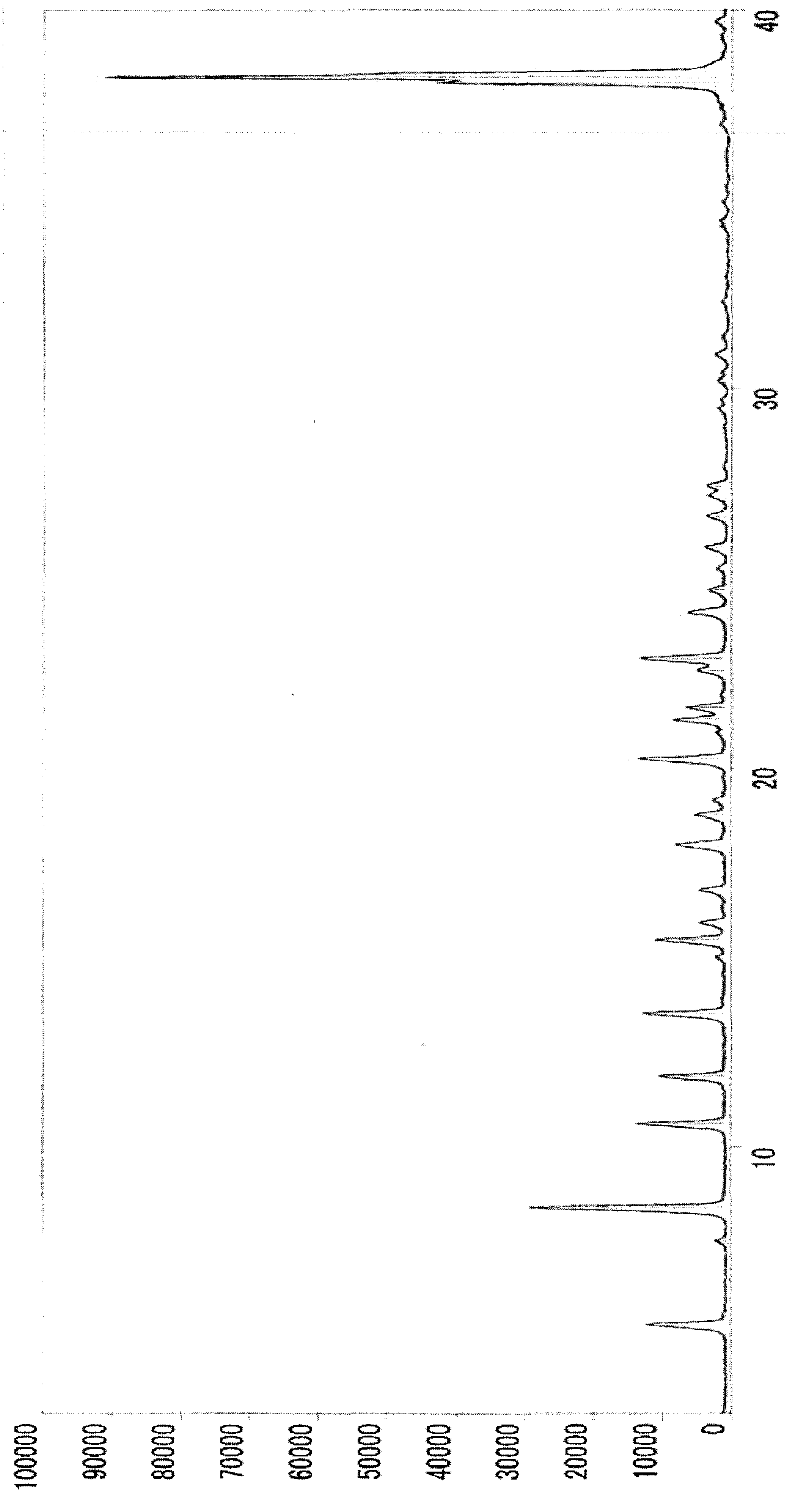
[0130] 1-Hexanol (22.5 g, 220 mmol) was added to compound 13 (12.0 g, 36.7 mmol) with stirring, and the temperature of the mixture was adjusted to 20°C. The above magnesium hexoxide solution was added to the resulting slurry of compound 13. The reaction mixture was stirred at 20°C for 4 hours, and then an aqueous solution of citric acid (3.1 g of citric acid monohydrate and 36 g of water) was added. The mixture was extracted with THF (10.7 g), and the organic layer was washed with water (24 g). The organic layer was concentrated to about 55 g under reduced pressure. A solution of p-toluenesulfonic acid in THF (7.0 g of p-toluenesulfonic acid monohydrate and 42.8 g of THF) was added to the resulting concentrate. The mixture was concentrated to about 61 g under reduced pressure. THF (42.7 g) was added to the concentrate, and the resulting mixture was concentrated to about 61 g under reduced pressure. After heating the mixture to 50°C, methyl tert-butyl ether (133.0 g) was added. The resulting mixture was cooled to 10°C, and stirred at 10°C for 1.5 hours. The resulting white precipitate was collected by filtration. The obtained solid was washed with a mixture of methyl tert-butyl ether (40.0 g) and ethyl acetate (16.0 g), and dried to obtain the tosylate of compound 20 (15.8 g, yield 87.2%) as white crystals.

[0131] ^{1}H-NMR($CDCl_3$) δ: 0.88 (t, J = 7.2 Hz, 3H), 1.25-1.34 (m, 4H), 1.34-1.43 (m, 2H), 1.76-1.85 (m, 2H), 2.34 (s, 3H), 3.04 (ddd, J = 13.6, 11.7, 4.3 Hz, 3H), 3.36 (dd, J = 11.6, 10.0 Hz, 3H), 3.43 (ddd, J = 13.6, 12.0, 4.4 Hz, 3H), 4.00 (dd, J = 11.7, 4.3 Hz, 1H), 4.06-4.18 (m, 4H), 4.80 (br, s, 1H), 7.16 (d, J = 7.8 Hz, 1H), 7.62 (d, J = 7.8 Hz, 1H), 7.62 (d, J = 7.1 Hz, 1H), 8.17 (d, J = 7.1 Hz, 1H), 8.40 (br, s, 1H).

[0132] Powder X-ray diffraction 2θ (°): 5.9, 8.4, 11.6, 12.7, 13.1, 15.7

[0133] The powder X-ray diffraction pattern of Compound 20 is shown in Fig. 4.

Fig. 1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